

Operating System

BCA — Semester 3 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. What are the different types of operating system? Differentiate between Real time and Batch OS.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What is kernel? Explain the process of system call with suitable diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What is Resource Allocation Graph (RAG)? What are the necessary conditions for deadlock? Explain. Determine if the system is safe from deadlock or not using Banker's Algorithm for given condition.[Note: Consider process sequence as: P1, P2, P3, P4, P5]Need MatrixAllocation MatrixMAX Resource MatrixProcessABCProcessABCABCP1143P131117810P2321P2131P3020P3411P4324P4010P5523P5201

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. What is an OS? Explain 3-state model of process with neat and clean diagram.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. What is race condition? Explain critical problem solution using lock variable.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is memory compaction? Explain best fit memory allocation with suitable example.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. What is RAID? Explain Level-2 and Level-3 RAID.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. How threads differ from processes? Explain user level thread and kernel level thread.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Consider the following page reference string: 3,2,1,3,4,2,3,9,7,4,2,9,8,3. Calculate the total number of page faults for Optimal and LRU page replacement algorithms using 4 page frame

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Explain the working mechanism of DMA.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. What is TLB? Explain the importance of TLB in conversion of logical address to physical address.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Consider 500 GB hard drive with 5 KB block size. Calculate the size of file allocation table if entry for each block needs 4 bytes.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.